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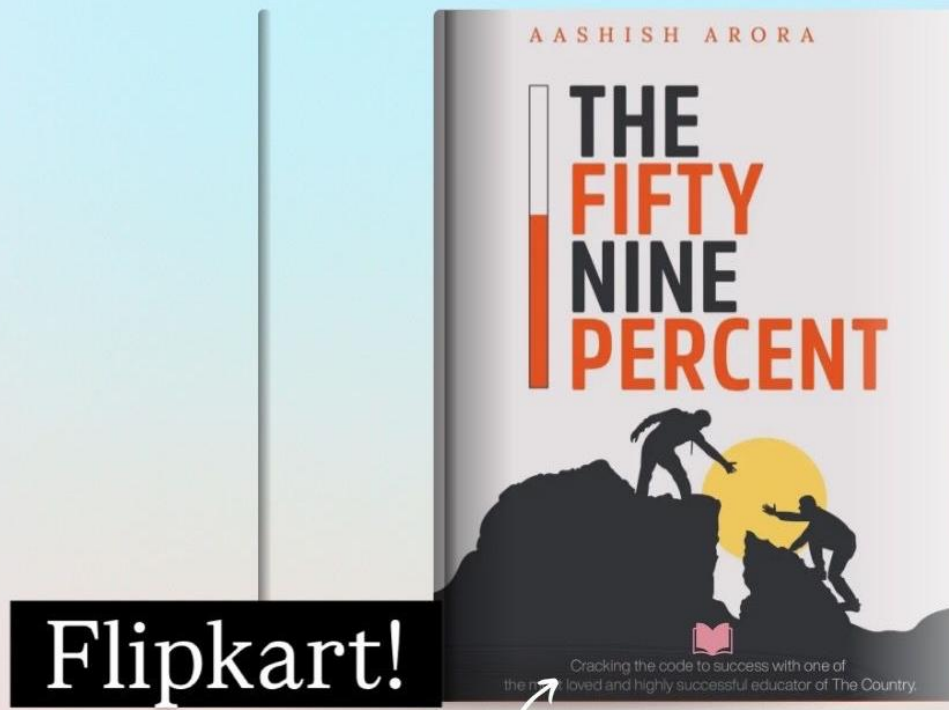


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Stay focused. Stay consistent. Work hard.

-Aashish Arora

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# 1. SIMPLIFICATION AND APPROXIMATION

**Direction: What value should come in place of the question mark (?) in the following question?**

1.  $37.5\% \text{ of } 864 + 14.28\% \text{ of } 735 + \sqrt{1681} = ? \times 8 + 52$

- A. 48.25
- B. 50.25
- C. 54.25
- D. 52.25
- E. None of these

2.  $68.75\% \text{ of } 512 - 31.25\% \text{ of } 448 + \sqrt[3]{15625} = ? + 225$

- A. 8
- B. 12
- C. 16
- D. 20
- E. None of these

3.  $[\frac{5}{12} \text{ of } 720 + \frac{7}{8} \text{ of } 256 - 18^2] = ? - 64$

- A. 252
- B. 264
- C. 276

D. 288

E. None of these

4.  $? \% \text{ of } 900 + 56.25\% \text{ of } 640 = 80\% \text{ of } 850 + 16 \times 25$

- A. 72
- B. 76
- C. 80
- D. 84
- E. None of these

5.  $\sqrt{7056} + 27.27\% \text{ of } 1210 + 2^3 \times 15 = ? \times 6 + 18$

- A. 86
- B. 82
- C. 90
- D. 78
- E. None of these

6.  $2\frac{3}{5} \text{ of } 150 + 11.11\% \text{ of } 1350 - 41.66\% \text{ of } 720 = ?$

- A. 210
- B. 220
- C. 230

D. 240

E. None of these

$$7. (24^2 - 16^2) \div 8 + 32.5\% \text{ of } 800 = ? \div 4$$

A. 1160

B. 1200

C. 1240

D. 1280

E. None of these

$$8. 75\% \text{ of } 960 \div 12 + 58.33\% \text{ of } 432 = ? + 210$$

A. 102

B. 98

C. 106

D. 110

E. None of these

$$9. 15^2 + \sqrt[3]{46656} - 18.75\% \text{ of } 704 = ? \times 5 - 64$$

A. 32.6

B. 34.6

C. 36.6

D. 38.6

E. None of these

$$10. \frac{7}{15} \text{ of } 675 + 6.25\% \text{ of } 960 + \sqrt{2704} = ? + 300$$

A. 122

B. 127

C. 132

D. 137

E. None of these

$$11. (42 \times 36) - (28 \times 44) + 83.33\% \text{ of } 672 = ? \times 7$$

A. 120

B. 116

C. 124

D. 128

E. None of these

$$12. 14400 \div \sqrt{576} \div 25 + 4^3 \times 6 = ? + 372$$

A. 28

B. 32

C. 40

D. 36

E. None of these

$$13. 71.42\% \text{ of } 882 + 44.44\% \text{ of } 405 - 22.22\% \text{ of } 810 = ? \times 9$$

A. 68

B. 70



C. 72

D. 74

E. None of these

14.  $3\frac{1}{4}$  of 128 + 25% of 936 - 68.75% of 320 = ?

A. 410

B. 420

C. 440

D. 430

E. None of these

15.  $? \times 18 - 18.75\% \text{ of } 1152 = \frac{5}{6} \text{ of } 864 + \sqrt{729}$

A. 51.5

B. 52.5

C. 53.5

D. 54.5

E. None of these

16.  $\sqrt{?} + 31.25\% \text{ of } 832 = 34^2 - 16^2$

A. 409600

B. 422500

C. 396900

D. 384400

E. None of these

17.  $(56.25\% \text{ of } 640 + \frac{2}{9} \text{ of } 1215) \div ? = \sqrt[3]{19683} + 15$

A. 12

B. 14

C. 15

D. 18

E. None of these

18.  $6.25 \times 48 + 12.5 \times 64 - 5.5 \times 72 = ?^2 + 448$

A. 14

B. 16

C. 18

D. 20

E. None of these

19.  $\frac{9}{16} \text{ of } 1600 + \frac{13}{40} \text{ of } 1200 - \frac{7}{12} \text{ of } 504 = ? + 710$

A. 276

B. 286

C. 296

D. 306

E. None of these

20.  $(\sqrt{9604} + \sqrt{5476}) \times 6 - 14^2 = ? \times 8$

A. 100.5

B. 102.5

- C. 106.5  
D. 104.5  
E. None of these

**Answers:**

1. D  
2. B  
3. B  
4. C  
5. A  
6. D  
7. B  
8. A  
9. D  
10. B  
11. A  
12. D  
13. B  
14. D  
15. C  
16. A  
17. C  
18. B  
19. B  
20. D

**Solutions:**

$$1. 37.5\% \text{ of } 864 + 14.28\% \text{ of } 735 + \sqrt{1681} = ? \times 8 + 52$$

$$37.5\% = \frac{3}{8}, 14.28\% = \frac{1}{7} \text{ and } \sqrt{1681} = 41$$

$$\frac{3}{8} \text{ of } 864 = 324 \text{ and } \frac{1}{7} \text{ of } 735 = 105$$

$$\text{So, } 324 + 105 + 41 = ? \times 8 + 52$$

$$470 = ? \times 8 + 52$$

$$? \times 8 = 418$$

$$? = 52.25$$

$$2. 68.75\% \text{ of } 512 - 31.25\% \text{ of } 448 + \sqrt[3]{15625} = ? + 225$$

$$68.75\% = \frac{11}{16} \text{ and } 31.25\% = \frac{5}{16}$$

$$\frac{11}{16} \text{ of } 512 = 352, \frac{5}{16} \text{ of } 448 = 140 \text{ and } \sqrt[3]{15625} = 25$$

$$\text{So, } 352 - 140 + 25 = ? + 225$$

$$237 = ? + 225$$

$$? = 12$$

$$3. \left[ \frac{5}{12} \text{ of } 720 + \frac{7}{8} \text{ of } 256 - 18^2 \right] = ? - 64$$

$$\frac{5}{12} \text{ of } 720 = 300, \frac{7}{8} \text{ of } 256 = 224 \text{ and } 18^2 = 324$$

$$\text{So, } 300 + 224 - 324 = ? - 64$$

$$200 = ? - 64$$

$$? = 264$$

$$4. ?\% \text{ of } 900 + 56.25\% \text{ of } 640 = 80\% \text{ of } 850 + 16 \times 25$$

$$56.25\% = \frac{9}{16} \text{ and } 80\% = \frac{4}{5}$$

$$\frac{9}{16} \text{ of } 640 = 360 \text{ and } \frac{4}{5} \text{ of } 850 = 680$$

$$?\% \text{ of } 900 + 360 = 680 + 400$$

$$?\% \text{ of } 900 + 360 = 1080$$

$$?\% \text{ of } 900 = 720$$

$$? = 720 \times 100 \div 900$$

$$? = 80$$

$$5. \sqrt{7056} + 27.27\% \text{ of } 1210 + 2^3 \times 15 = ? \times 6 + 18$$

$$\sqrt{7056} = 84, 27.27\% = \frac{3}{11} \text{ and } 2^3 = 8$$

$$\frac{3}{11} \text{ of } 1210 = 330$$

$$\text{So, } 84 + 330 + 8 \times 15 = ? \times 6 + 18$$

$$84 + 330 + 120 = ? \times 6 + 18$$

$$534 = ? \times 6 + 18$$

$$? \times 6 = 516$$

$$? = 86$$

$$6. 2\frac{3}{5} \text{ of } 150 + 11.11\% \text{ of } 1350 - 41.66\% \text{ of } 720 = ?$$

$$2\frac{3}{5} = \frac{13}{5}, 11.11\% = \frac{1}{9} \text{ and } 41.66\% = \frac{5}{12}$$

$$\frac{13}{5} \text{ of } 150 = 390, \frac{1}{9} \text{ of } 1350 = 150 \text{ and } \frac{5}{12} \text{ of } 720 = 300$$

$$\text{So, } 390 + 150 - 300 = ?$$

$$? = 240$$

$$7. (24^2 - 16^2) \div 8 + 32.5\% \text{ of } 800 = ? \div 4$$

$$24^2 = 576 \text{ and } 16^2 = 256$$

$$32.5\% = \frac{13}{40} \text{ and } \frac{13}{40} \text{ of } 800 = 260$$

$$\text{So, } (576 - 256) \div 8 + 260 = ? \div 4$$

$$320 \div 8 + 260 = ? \div 4$$

$$40 + 260 = ? \div 4$$

$$300 = ? \div 4$$

$$? = 1200$$

$$8. 75\% \text{ of } 960 \div 12 + 58.33\% \text{ of } 432 = ? + 210$$

$$75\% = \frac{3}{4} \text{ and } 58.33\% = \frac{7}{12}$$

$$\frac{3}{4} \text{ of } 960 = 720 \text{ and } \frac{7}{12} \text{ of } 432 = 252$$

$$\text{So, } 720 \div 12 + 252 = ? + 210$$

$$60 + 252 = ? + 210$$

$$312 = ? + 210$$

$$? = 102$$

$$9. 15^2 + \sqrt[3]{46656} - 18.75\% \text{ of } 704 = ? \times 5 - 64$$

$$15^2 = 225, \sqrt[3]{46656} = 36 \text{ and } 18.75\% = \frac{3}{16}$$

$$\frac{3}{16} \text{ of } 704 = 132$$

$$\text{So, } 225 + 36 - 132 = ? \times 5 - 64$$

$$129 = ? \times 5 - 64$$

$$? \times 5 = 193$$

$$? = 38.6$$

$$10. \frac{7}{15} \text{ of } 675 + 6.25\% \text{ of } 960 + \sqrt{2704} = ? + 300$$

$$\frac{7}{15} \text{ of } 675 = 315, 6.25\% = \frac{1}{16} \text{ and } \sqrt{2704} = 52$$

$$\frac{1}{16} \text{ of } 960 = 60$$

$$\text{So, } 315 + 60 + 52 = ? + 300$$

$$427 = ? + 300$$

$$? = 127$$

$$11. (42 \times 36) - (28 \times 44) + 83.33\% \text{ of } 672 = ? \times 7$$

$$42 \times 36 = 1512 \text{ and } 28 \times 44 = 1232$$

$$83.33\% = \frac{5}{6} \text{ and } \frac{5}{6} \text{ of } 672 = 560$$

$$\text{So, } 1512 - 1232 + 560 = ? \times 7$$

$$840 = ? \times 7$$

$$? = 120$$

$$12. 14400 \div \sqrt{576} \div 25 + 4^3 \times 6 = ? + 372$$

$$\sqrt{576} = 24 \text{ and } 4^3 = 64$$

$$14400 \div 24 \div 25 = 24 \text{ and } 64 \times 6 = 384$$

$$\text{So, } 24 + 384 = ? + 372$$

$$408 = ? + 372$$

$$? = 36$$

$$13. 71.42\% \text{ of } 882 + 44.44\% \text{ of } 405 - 22.22\% \text{ of } 810 = ? \times 9$$

$$71.42\% = \frac{5}{7}, 44.44\% = \frac{4}{9} \text{ and } 22.22\% = \frac{2}{9}$$

$$\frac{5}{7} \text{ of } 882 = 630, \frac{4}{9} \text{ of } 405 = 180 \text{ and } \frac{2}{9} \text{ of } 810 = 180$$

$$\text{So, } 630 + 180 - 180 = ? \times 9$$

$$630 = ? \times 9$$

$$? = 70$$

$$14. 3\frac{1}{4} \text{ of } 128 + 25\% \text{ of } 936 - 68.75\% \text{ of } 320 = ?$$

$$3\frac{1}{4} = \frac{13}{4}, 25\% = \frac{1}{4} \text{ and } 68.75\% = \frac{11}{16}$$

$$\frac{13}{4} \text{ of } 128 = 416, \frac{1}{4} \text{ of } 936 = 234 \text{ and } \frac{11}{16} \text{ of } 320 = 220$$

$$\text{So, } 416 + 234 - 220 = ?$$

$$? = 430$$

$$15. ? \times 18 - 18.75\% \text{ of } 1152 = \frac{5}{6} \text{ of } 864 + \sqrt{729}$$

$$18.75\% = \frac{3}{16} \text{ and } \sqrt{729} = 27$$

$$\frac{3}{16} \text{ of } 1152 = 216 \text{ and } \frac{5}{6} \text{ of } 864 = 720$$

$$\text{So, } ? \times 18 - 216 = 720 + 27$$

$$? \times 18 - 216 = 747$$

$$? \times 18 = 963$$

$$? = 53.5$$

$$16. \sqrt{?} + 31.25\% \text{ of } 832 = 34^2 - 16^2$$

$$31.25\% = \frac{5}{16}, 34^2 = 1156 \text{ and } 16^2 = 256$$

$$\frac{5}{16} \text{ of } 832 = 260$$

$$\text{So, } \sqrt{?} + 260 = 1156 - 256$$

$$\sqrt{?} + 260 = 900$$

$$\sqrt{?} = 640$$

$$? = 640^2$$

$$? = 409600$$

$$17. (56.25\% \text{ of } 640 + \frac{2}{9} \text{ of } 1215) \div ? = \sqrt[3]{19683} + 15$$

$$56.25\% = \frac{9}{16} \text{ and } \sqrt[3]{19683} = 27$$

$$\frac{9}{16} \text{ of } 640 = 360 \text{ and } \frac{2}{9} \text{ of } 1215 = 270$$

$$\text{So, } (360 + 270) \div ? = 27 + 15$$

$$630 \div ? = 42$$

$$? = 630 \div 42$$

$$? = 15$$

$$18. 6.25 \times 48 + 12.5 \times 64 - 5.5 \times 72 = ?^2 + 448$$

$$6.25 \times 48 = 300, 12.5 \times 64 = 800 \text{ and } 5.5 \times 72 = 396$$

$$\text{So, } 300 + 800 - 396 = ?^2 + 448$$

$$704 = ?^2 + 448$$

$$?^2 = 256$$

$$? = 16$$

$$19. \frac{9}{16} \text{ of } 1600 + \frac{13}{40} \text{ of } 1200 - \frac{7}{12} \text{ of } 504 = ? + 710$$

$$\frac{9}{16} \text{ of } 1600 = 900, \frac{13}{40} \text{ of } 1200 = 390$$

$$\text{and } \frac{7}{12} \text{ of } 504 = 294$$

$$\text{So, } 900 + 390 - 294 = ? + 710$$

$$996 = ? + 710$$

$$? = 286$$

$$20. (\sqrt{9604} + \sqrt{5476}) \times 6 - 14^2 = ? \times 8$$

$$\sqrt{9604} = 98, \sqrt{5476} = 74 \text{ and } 14^2 = 196$$

$$\text{So, } (98 + 74) \times 6 - 196 = ? \times 8$$

$$172 \times 6 - 196 = ? \times 8$$

$$1032 - 196 = ? \times 8$$

$$836 = ? \times 8$$

$$? = 104.5$$



## 2. ARITHMETIC QUESTIONS

Q1: The present ages of Advik and Bhavesh are in the ratio of 5:4 respectively. If the ratio of Advik's age 8 years hence to Bhavesh's age 2 years ago is 3:2, then find the sum of their ages 6 years hence.

अद्विक और भावेश की वर्तमान आयु का अनुपात क्रमशः 5:4 है। यदि 8 वर्ष बाद अद्विक की आयु और 2 वर्ष पहले भावेश की आयु का अनुपात 3:2 है, तो 6 वर्ष बाद उनकी आयु का योग ज्ञात कीजिए।

- (A) 101 years
- (B) 111 years
- (C) 99 years
- (D) 121 years
- (E) 108 years

Q2: A teacher brought X pencils to distribute equally among 40 students. However, 8 students were absent, so each of the remaining students received 3 pencils more than they would have received if all 40 students were present. Find X.

एक शिक्षक 40 विद्यार्थियों में बराबर बाँटने के लिए x पेंसिल लाया। लेकिन 8 विद्यार्थी अनुपस्थित थे, इसलिए शेष प्रत्येक विद्यार्थी को सभी 40 विद्यार्थियों के उपस्थित होने की तुलना में 3 पेंसिल अधिक मिलीं। x ज्ञात कीजिए।

- (A) 360
- (B) 420
- (C) 500
- (D) 480
- (E) 600

Q3: Two positive integers P and Q are in the ratio of 5:3 respectively. If 20 is subtracted from P and added to Q, then the final ratio of P to Q becomes 5:7. Find the initial value of P.

दो धनात्मक पूर्णांक P और Q क्रमशः 5:3 के अनुपात में हैं। यदि P में से 20 घटाकर Q में जोड़ दिया जाए, तो P और Q का अंतिम अनुपात 5:7 हो जाता है। P का प्रारंभिक मान ज्ञात कीजिए।

- (A) 48

- (B) 72
- (C) 60
- (D) 54
- (E) 66

Q4: The total number of adults and children who visited a museum on Sunday was 72. The ticket price for each adult was Rs. 45 and for each child was Rs. 25. If the total amount collected was Rs. 2500, find the number of adults.

रविवार को संग्रहालय देखने आए वयस्कों और बच्चों की कुल संख्या 72 थी। प्रत्येक वयस्क का टिकट मूल्य Rs. 45 और प्रत्येक बच्चे का टिकट मूल्य Rs. 25 था। यदि कुल संग्रह Rs. 2500 था, तो वयस्कों की संख्या ज्ञात कीजिए।

- (A) 30
- (B) 25
- (C) 40
- (D) 32
- (E) 35

Q5: A shopkeeper marked article P 40% above its cost price and allowed a discount of 15%. He earned a profit of Rs. 171 on article P. The cost price of article Q is 30% more than that of article P. If article Q is sold at a loss of 20%, find the selling price of article Q.

एक दुकानदार ने वस्तु P का अंकित मूल्य उसके क्रय मूल्य से 40% अधिक रखा और 15% की छूट दी। वस्तु P पर उसे Rs. 171 का लाभ हुआ। वस्तु Q का क्रय मूल्य वस्तु P के क्रय मूल्य से 30% अधिक है। यदि वस्तु Q को 20% हानि पर बेचा जाता है, तो वस्तु Q का विक्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 880
- (B) Rs. 936
- (C) Rs. 960
- (D) Rs. 920
- (E) Rs. 900

Q6: The length of Train P is  $12x$  meters, and the length of Train Q is one-third more than the length of Train P. The two trains cross each other in 16 seconds while running in opposite directions. The speed of Train P is 15 meters per second, and the speed of Train Q is 72 kilometers per hour. Find the time taken by Train Q to cross a bridge whose length is equal to one-fourth of the length of Train P.

ट्रेन P की लंबाई 12x मीटर है और ट्रेन Q की लंबाई ट्रेन P की लंबाई से एक-तिहाई अधिक है। दोनों ट्रेनें विपरीत दिशाओं में चलते हुए 16 सेकंड में एक-दूसरे को पार करती हैं। ट्रेन P की गति 15 मीटर प्रति सेकंड है और ट्रेन Q की गति 72 किलोमीटर प्रति घंटा है। ट्रेन Q को उस पुल को पार करने में कितना समय लगेगा जिसकी लंबाई ट्रेन P की लंबाई के एक-चौथाई के बराबर है?

- (A) 16 seconds
- (B) 18 seconds
- (C) 19 seconds
- (D) 21 seconds
- (E) 24 seconds

Q7: A boat covers Y km upstream in 40 minutes and 3Y km downstream in 1 hour. Find the speed of the boat in still water as a percentage of the speed of the stream.

एक नाव 40 मिनट में धारा के प्रतिकूल Y किमी और 1 घंटे में धारा के अनुकूल 3Y किमी की दूरी तय करती है। स्थिर जल में नाव की गति, धारा की गति का कितना प्रतिशत है?

- (A) 250%
- (B) 400%
- (C) 200%
- (D) 300%
- (E) 350%

Q8: Two rectangular gardens have equal areas. The length of the first garden is p% less than the length of the second garden, and the breadth of the first garden is 5p% greater than the breadth of the second garden. Find the value of p, where p is a non-zero number.

दो आयताकार बगीचों के क्षेत्रफल समान हैं। पहले बगीचे की लंबाई दूसरे बगीचे की लंबाई से p% कम है और पहले बगीचे की चौड़ाई दूसरे बगीचे की चौड़ाई से 5p% अधिक है। p का मान ज्ञात कीजिए, जहां p शून्य नहीं है।

- (A) 60
- (B) 75
- (C) 50
- (D) 90
- (E) 80

Q9: Himanshu can complete a work in 30 days. Praveen is three times as efficient as Himanshu. The efficiency ratio of Sonal to Praveen is 2 : 3. If Himanshu,

Praveen and Sonal work together, in how many days will the work be completed?  
 हिमांशु किसी काम को 30 दिनों में पूरा कर सकता है। प्रवीण, हिमांशु से तीन गुना दक्ष है।  
 सोनल और प्रवीण की दक्षता का अनुपात 2 : 3 है। यदि हिमांशु, प्रवीण और सोनल मिलकर  
 काम करें, तो काम कितने दिनों में पूरा होगा?

- (A) 4 days
- (B) 5 days
- (C) 6 days
- (D) 8 days
- (E) 10 days

Q10: Mahesh invested Rs. Y in a business. After 5 months, Nilesh joined the business with Rs. 8000. After another 2 months, Mahesh withdrew Rs. 2000. At the end of the year, Mahesh's share of the profit was Rs. 7800 out of a total profit of Rs. 12000. Find the value of Y.

महेश ने एक व्यवसाय में Rs. Y निवेश किए। 5 महीने बाद, निलेश Rs. 8000 के साथ व्यवसाय में शामिल हुआ। अगले 2 महीने बाद, महेश ने Rs. 2000 निकाल लिए। वर्ष के अंत में, कुल Rs. 12000 के लाभ में से महेश का हिस्सा Rs. 7800 था। Y का मान ज्ञात कीजिए।

- (A) Rs. 9000
- (B) Rs. 9500
- (C) Rs. 10000
- (D) Rs. 8500
- (E) Rs. 11000

Q11: Harsh, Isha and Ketan have money in the ratio 4:6:7 respectively. If Rs 450 is distributed equally among them, Ketan will have Rs 850. Find the original amount with Harsh.

हर्ष, ईशा और केतन के पास धन का अनुपात क्रमशः 4:6:7 है। यदि रु. 450 को उनके बीच बराबर बांटा जाए, तो केतन के पास रु. 850 हो जाएंगे। हर्ष के पास प्रारंभिक राशि ज्ञात कीजिए।

- (A) 350
- (B) 400
- (C) 450
- (D) 500
- (E) 550

Q12: The ratio of the speed of a boat in still water to the speed of the stream is 5:2. The time taken by the boat to cover  $(T + 84)$  km downstream is equal to the

time taken to cover  $(T - 60)$  km upstream. Find the value of  $T$ .

शांत जल में नाव की गति और धारा की गति का अनुपात 5:2 है। नाव द्वारा धारा के अनुकूल  $(T + 84)$  किमी की दूरी तय करने में लगा समय, धारा के प्रतिकूल  $(T - 60)$  किमी की दूरी तय करने में लगे समय के बराबर है।  $T$  का मान ज्ञात कीजिए।

- (A) 156
- (B) 180
- (C) 144
- (D) 168
- (E) 192

Q13: Sagar alone can complete a work in 24 days, and Ritu alone can complete the same work in 36 days. Sagar works every day from the start till completion. Ritu joins only for the last  $k$  days. If the work is completed in 20 days, find the value of  $k$ .

सागर अकेले एक कार्य को 24 दिनों में पूरा कर सकता है, और रितु अकेले उसी कार्य को 36 दिनों में पूरा कर सकती है। सागर शुरुआत से अंत तक हर दिन काम करता है। रितु केवल अंतिम  $k$  दिनों के लिए काम में शामिल होती है। यदि कार्य 20 दिनों में पूरा होता है, तो  $k$  का मान ज्ञात कीजिए।

- (A) 4
- (B) 5
- (C) 8
- (D) 9
- (E) 6

Q14: The average score of students in Section A is 52, and the average score of students in Section B is 64. If the average score of all students in both sections together is 57, find the sum of the terms of the ratio of the number of students in Section A to that in Section B.

सेक्शन A के छात्रों का औसत अंक 52 है, और सेक्शन B के छात्रों का औसत अंक 64 है। यदि दोनों सेक्शन के सभी छात्रों का कुल औसत अंक 57 है, तो सेक्शन A और सेक्शन B में छात्रों की संख्या के अनुपात के पदों का योग ज्ञात कीजिए।

- (A) 12
- (B) 10
- (C) 14
- (D) 15
- (E) 11

Q15: A tank is full of a mixture of milk and water in which water is 24%. If 9 liters of the mixture is drawn out and replaced with pure milk, the percentage of water becomes 18%. Find the capacity of the tank.

एक टंकी दूध और पानी के मिश्रण से भरी है, जिसमें पानी 24% है। यदि 9 लीटर मिश्रण निकालकर उसकी जगह शुद्ध दूध डाल दिया जाए, तो पानी का प्रतिशत 18% हो जाता है। टंकी की क्षमता ज्ञात कीजिए।

- (A) 30 liters
- (B) 42 liters
- (C) 36 liters
- (D) 45 liters
- (E) 48 liters

Q16: A boatman can row at 10 km/hr in still water. The speed of the stream is 2 km/hr. If he takes 3 hours more to cover a certain distance upstream than downstream, find the distance.

एक नाविक स्थिर जल में 10 किमी/घंटा की गति से नाव चला सकता है। धारा की गति 2 किमी/घंटा है। यदि वह एक निश्चित दूरी को धारा के विपरीत तय करने में धारा के अनुकूल की तुलना में 3 घंटे अधिक लेता है, तो दूरी ज्ञात कीजिए।

- (A) 54 km
- (B) 60 km
- (C) 72 km
- (D) 84 km
- (E) 96 km

Q17: The ratio of the volume of a cylinder to the volume of a sphere is 9 : 2. If the radius of the cylinder is equal to the radius of the sphere, find the ratio of the total surface area of the cylinder to the total surface area of the sphere.

एक बेलन के आयतन और एक गोले के आयतन का अनुपात 9 : 2 है। यदि बेलन की त्रिज्या गोले की त्रिज्या के बराबर है, तो बेलन के कुल पृष्ठीय क्षेत्रफल और गोले के कुल पृष्ठीय क्षेत्रफल का अनुपात ज्ञात कीजिए।

- (A) 5 : 2
- (B) 7 : 2
- (C) 3 : 1
- (D) 4 : 1
- (E) 9 : 4



Q18: A person's income is Rs. 24000, and he saves Rs.  $x$ . If his income becomes Rs. 42000, then by what percent should his savings increase so that his savings percentage remains unchanged?

एक व्यक्ति की आय Rs. 24000 है और वह Rs.  $x$  बचाता है। यदि उसकी आय Rs. 42000 हो जाती है, तो उसकी बचत कितने प्रतिशत बढ़नी चाहिए ताकि उसकी बचत प्रतिशत में कोई बदलाव न हो?

- (A) 60%
- (B) 65%
- (C) 70%
- (D) 75%
- (E) 80%

Q19: Worker P can complete a task in 10 hours, and worker Q can complete the same task in 15 hours. Find the time taken by P and Q together to complete the task.

मजदूर P एक कार्य को 10 घंटे में पूरा कर सकता है और मजदूर Q उसी कार्य को 15 घंटे में पूरा कर सकता है। P और Q मिलकर उस कार्य को पूरा करने में कितना समय लेंगे?

- (A) 4 hours
- (B) 5 hours
- (C) 6 hours
- (D) 8 hours
- (E) 9 hours

Q20: Two trains P and Q are 300 meters and 450 meters long respectively. Train P runs at 54 km/hr, and Train Q runs at 25 m/sec. Find the ratio of the time taken by them to cross each other when they run in opposite directions to the time taken when they run in the same direction.

दो ट्रेन P और Q की लंबाई क्रमशः 300 मीटर और 450 मीटर है। ट्रेन P की गति 54 किमी/घंटा है और ट्रेन Q की गति 25 मीटर/सेकंड है। जब वे विपरीत दिशाओं में चलती हैं और जब वे समान दिशा में चलती हैं, तब एक-दूसरे को पार करने में लगने वाले समय का अनुपात ज्ञात कीजिए।

- (A) 1 : 2
- (B) 1 : 3
- (C) 2 : 3
- (D) 1 : 4
- (E) 3 : 4

**Answers:**

- |      |      |
|------|------|
| 1. B | 11.B |
| 2. D | 12.D |
| 3. C | 13.E |
| 4. E | 14.A |
| 5. B | 15.C |
| 6. C | 16.C |
| 7. D | 17.B |
| 8. E | 18.D |
| 9. B | 19.C |
| 10.B | 20.D |

**Solutions:**

1. Option B

Let Advik's present age be  $5x$  years.

Let Bhavesh's present age be  $4x$  years.

Advik's age 8 years hence =  $5x + 8$

Bhavesh's age 2 years ago =  $4x - 2$

Given ratio = 3:2

So:

$$(5x + 8) : (4x - 2) = 3 : 2$$

Now cross multiply.

$$2(5x + 8) = 3(4x - 2)$$

$$10x + 16 = 12x - 6$$

$$12x - 10x = 16 + 6$$

$$2x = 22$$

$$x = 22/2$$

$$x = 11$$

Advik's present age =  $5 \times 11 = 55$  years

Bhavesh's present age =  $4 \times 11 = 44$  years

Advik's age 6 years hence =  $55 + 6 = 61$  years

Bhavesh's age 6 years hence =  $44 + 6 = 50$  years

Required sum =  $61 + 50 = 111$  years

## 2. Option D

Total number of students = 40

Absent students = 8

Present students =  $40 - 8 = 32$

If all 40 students were present, each student would get  $X/40$  pencils.

When only 32 students were present, each student got  $X/32$  pencils.

Each present student received 3 pencils more.

So:

$$X/32 - X/40 = 3$$

LCM of 32 and 40 = 160

$$5X/160 - 4X/160 = 3$$

$$X/160 = 3$$

$$X = 3 \times 160$$

$$X = 480$$

## 3. Option C

Let the initial values of P and Q be  $5x$  and  $3x$  respectively.

20 is subtracted from P.

New value of P =  $5x - 20$

20 is added to Q.

New value of Q =  $3x + 20$

Final ratio = 5:7

So:

$$(5x - 20) : (3x + 20) = 5 : 7$$

Now cross multiply.

$$7(5x - 20) = 5(3x + 20)$$

$$35x - 140 = 15x + 100$$

$$35x - 15x = 100 + 140$$

$$20x = 240$$

$$x = 240/20$$

$$x = 12$$

Initial value of P =  $5x$

Initial value of P =  $5 \times 12 = 60$

#### 4. Option E

Let the number of adults be  $x$ .

Total number of adults and children = 72

So, number of children =  $72 - x$

Ticket price for each adult = Rs. 45

Ticket price for each child = Rs. 25

Total amount collected = Rs. 2500

So:

$$45x + 25(72 - x) = 2500$$

$$45x + 1800 - 25x = 2500$$

$$20x + 1800 = 2500$$

$$20x = 2500 - 1800$$

$$20x = 700$$

$$x = 700/20$$

$$x = 35$$

#### 5. Option B

Let the cost price of article P be Rs.  $x$ .

Marked price of article P = 140% of  $x$

Marked price =  $140x/100$

Discount = 15%

Selling price of article P = 85% of marked price

Selling price of article P =  $85/100 \times 140x/100$

Selling price of article P =  $119x/100$

Profit on article P = Rs. 171

Profit = Selling price - Cost price

$$119x/100 - x = 171$$

$$119x/100 - 100x/100 = 171$$

$$19x/100 = 171$$

$$x = 171 \times 100/19$$

$$x = 900$$

Cost price of article P = Rs. 900

Cost price of article Q is 30% more than article P.

Cost price of article Q = 130% of 900

Cost price of article Q =  $130 \times 900/100$

Cost price of article Q = Rs. 1170

Article Q is sold at 20% loss.

Selling price of article Q = 80% of 1170

Selling price of article Q =  $80 \times 1170/100$

Selling price of article Q = Rs. 936

#### 6. Option C

Speed of Train P = 15 m/sec.

Speed of Train Q = 72 km/hr.

Convert 72 km/hr into m/sec.

Speed of Train Q =  $72 \times 5/18 = 20$  m/sec.

Length of Train P =  $12x$  meters.

Train Q is one-third more than Train P.

Length of Train Q =  $12x + 1/3 \times 12x = 12x + 4x = 16x$  meters.

Both trains run in opposite directions.

Relative speed =  $15 + 20 = 35$  m/sec.

They cross each other in 16 seconds.

Total distance covered =  $35 \times 16 = 560$  meters.

So,

$$12x + 16x = 560$$

$$28x = 560$$

$$x = 560/28$$

$$x = 20$$

Length of Train P =  $12 \times 20 = 240$  meters.

Length of Train Q =  $16 \times 20 = 320$  meters.

Length of bridge =  $1/4$  of Train P.

Length of bridge =  $240/4 = 60$  meters.

Train Q has to cross the bridge.

Total distance covered by Train Q =  $320 + 60 = 380$  meters.

Speed of Train Q = 20 m/sec.

Time taken =  $380/20 = 19$  seconds.

#### 7. Option D

Upstream distance =  $Y$  km.

Upstream time = 40 minutes =  $40/60$  hours =  $2/3$  hours.

Upstream speed = Distance/Time.

Upstream speed =  $Y/(2/3) = 3Y/2$  km/hr.

Downstream distance =  $3Y$  km.

Downstream time = 1 hour.

Downstream speed =  $3Y/1 = 3Y$  km/hr.

Speed of boat in still water = (Downstream speed + Upstream speed)/2.

Speed of boat in still water =  $(3Y + 3Y/2)/2$ .

Speed of boat in still water =  $(9Y/2)/2 = 9Y/4$ .

Speed of stream = (Downstream speed - Upstream speed)/2.

Speed of stream =  $(3Y - 3Y/2)/2$ .

Speed of stream =  $(3Y/2)/2 = 3Y/4$ .

Required percentage = (Speed of boat in still water/Speed of stream)  $\times$  100.

Required percentage =  $(9Y/4)/(3Y/4) \times 100$ .

Required percentage =  $3 \times 100 = 300\%$ .

#### 8. Option E

Let the length and breadth of the second garden be L and B.

Length of the first garden is p% less than the second garden.

Length of first garden =  $L \times (100 - p)/100$ .

Breadth of the first garden is 5p% greater than the second garden.

Breadth of first garden =  $B \times (100 + 5p)/100$ .

Both gardens have equal areas.

So,

$L \times B = L \times (100 - p)/100 \times B \times (100 + 5p)/100$

Cancel  $L \times B$  from both sides.

$1 = (100 - p)(100 + 5p)/10000$

So,

$(100 - p)(100 + 5p) = 10000$

$10000 + 500p - 100p - 5p^2 = 10000$

$400p - 5p^2 = 0$

$5p(80 - p) = 0$

$p = 0$  or  $p = 80$ .

Given p is a non-zero number.

So,  $p = 80$ .

#### 9. Option B

Take Himanshu's efficiency as 1 unit per day.

Himanshu can complete the work in 30 days.

So, total work = 30 units.

Praveen is three times as efficient as Himanshu.

Praveen's efficiency = 3 units per day.

The efficiency ratio of Sonal to Praveen is 2 : 3.

Since Praveen's efficiency = 3 units per day, Sonal's efficiency = 2 units per day.

Total efficiency of all three =  $1 + 3 + 2 = 6$  units per day.

Total work = 30 units.

Time taken =  $30/6 = 5$  days.

Therefore, they will complete the work in 5 days.

#### 10. Option B

Total profit = Rs. 12000.

Mahesh's profit share = Rs. 7800.

Nilesh's profit share =  $12000 - 7800 = \text{Rs. } 4200$ .

Profit ratio of Mahesh and Nilesh = 7800 : 4200.

Profit ratio = 13 : 7.

Profit depends on investment  $\times$  time.

Mahesh invested Rs. Y for the first 7 months.

After that, he withdrew Rs. 2000.

So, Mahesh's investment for the remaining 5 months = Rs.  $(Y - 2000)$ .

Mahesh's total investment value =  $Y \times 7 + (Y - 2000) \times 5$ .

Mahesh's total investment value =  $7Y + 5Y - 10000 = 12Y - 10000$ .

Nilesh invested Rs. 8000 after 5 months.

So, Nilesh invested for 7 months.

Nilesh's total investment value =  $8000 \times 7 = 56000$ .

Now,

$(12Y - 10000) : 56000 = 13 : 7$

Using cross multiplication:

$7(12Y - 10000) = 13 \times 56000$

$84Y - 70000 = 728000$

$84Y = 728000 + 70000$

$84Y = 798000$

$Y = 798000/84$

$Y = 9500$

#### 11. Option B

Total amount distributed equally = Rs 450.

Number of persons = 3.

Amount received by each person =  $450/3 = \text{Rs } 150$ .



After receiving Rs 150, Ketan has Rs 850.

So, Ketan's original amount =  $850 - 150 = \text{Rs } 700$ .

Money ratio of Harsh, Isha and Ketan = 4:6:7.

Ketan's 7 parts = Rs 700.

So, 1 part =  $700/7 = \text{Rs } 100$ .

Harsh's original amount = 4 parts.

Harsh's original amount =  $4 \times 100 = \text{Rs } 400$ .

## 12. Option D

Let the speed of the boat in still water be  $5x$  and the speed of the stream be  $2x$ .

Downstream speed =  $5x + 2x = 7x$ .

Upstream speed =  $5x - 2x = 3x$ .

Time taken downstream =  $(T + 84)/7x$ .

Time taken upstream =  $(T - 60)/3x$ .

Given, both times are equal.

So,  $(T + 84)/7x = (T - 60)/3x$ .

Cancel  $x$  from both sides.

$(T + 84)/7 = (T - 60)/3$ .

Cross multiply:

$3 \times (T + 84) = 7 \times (T - 60)$

$3T + 252 = 7T - 420$

$252 + 420 = 7T - 3T$

$672 = 4T$

$T = 672/4$

$T = 168$ .

## 13. Option E

Take total work as 72 units, because 72 is divisible by 24 and 36.

Sagar can complete the work in 24 days.

Sagar's 1 day work =  $72/24 = 3$  units.

Ritu can complete the work in 36 days.

Ritu's 1 day work =  $72/36 = 2$  units.

Sagar works for all 20 days.

Work done by Sagar in 20 days =  $20 \times 3 = 60$  units.

Remaining work =  $72 - 60 = 12$  units.

This remaining work is done by Ritu in the last  $k$  days.

Ritu does 2 units in 1 day.

So,  $k = 12/2 = 6$ .

#### 14. Option A

Average score of Section A = 52.

Average score of Section B = 64.

Combined average score = 57.

Using alligation:

Number of students in Section A : Number of students in Section B =  $(64 - 57) : (57 - 52)$ .

So, ratio = 7 : 5.

Sum of the terms of the ratio =  $7 + 5 = 12$ .

#### 15. Option C

Let the capacity of the tank be  $x$  liters.

Initial water percentage = 24%.

Initial quantity of water =  $24x/100$ .

9 liters of mixture is removed.

So, fraction of mixture removed =  $9/x$ .

After removing 9 liters and replacing it with pure milk, only water decreases.

Final water quantity =  $24x/100 \times (1 - 9/x)$ .

Final water percentage becomes 18%.

So, final water quantity =  $18x/100$ .

Now,

$$24x/100 \times (1 - 9/x) = 18x/100$$

Cancel  $x/100$  from both sides.

$$24 \times (1 - 9/x) = 18$$

$$1 - 9/x = 18/24$$

$$1 - 9/x = 3/4$$

$$9/x = 1 - 3/4$$

$$9/x = 1/4$$

$$x = 9 \times 4$$

$$x = 36.$$

#### 16. Option C

Speed of boat in still water = 10 km/hr

Speed of stream = 2 km/hr

Downstream speed =  $10 + 2 = 12$  km/hr

Upstream speed =  $10 - 2 = 8$  km/hr

Let the distance be  $D$  km.

Time taken upstream =  $D/8$  hours

Time taken downstream =  $D/12$  hours

Given, upstream time is 3 hours more than downstream time.

So,  $D/8 - D/12 = 3$

LCM of 8 and 12 = 24

$3D/24 - 2D/24 = 3$

$D/24 = 3$

$D = 3 \times 24 = 72$

#### 17. Option B

Let the common radius of the cylinder and sphere be  $r$ .

Let the height of the cylinder be  $h$ .

Volume of cylinder =  $\pi r^2 h$

Volume of sphere =  $4/3 \times \pi r^3$

Given, volume of cylinder : volume of sphere =  $9 : 2$

So,  $\pi r^2 h / (4/3 \times \pi r^3) = 9/2$

$3h/4r = 9/2$

Cross multiply:

$6h = 36r$

$h = 6r$

Now, total surface area of cylinder =  $2\pi r(h + r)$

Total surface area of sphere =  $4\pi r^2$

Required ratio =  $2\pi r(h + r) : 4\pi r^2$

Put  $h = 6r$ .

Ratio =  $2\pi r(6r + r) : 4\pi r^2$

Ratio =  $14\pi r^2 : 4\pi r^2$

Ratio =  $14 : 4 = 7 : 2$

#### 18. Option D

To keep the savings percentage unchanged, savings must increase in the same ratio as income.

Initial income = Rs. 24000

New income = Rs. 42000

Increase in income =  $42000 - 24000 = \text{Rs. } 18000$

Percentage increase in income =  $18000 \times 100/24000$

Percentage increase = 75%

So, savings should also increase by 75%.

### 19. Option C

P can complete the task in 10 hours.

So, P's one-hour work =  $\frac{1}{10}$

Q can complete the task in 15 hours.

So, Q's one-hour work =  $\frac{1}{15}$

Together, one-hour work of P and Q =  $\frac{1}{10} + \frac{1}{15}$

LCM of 10 and 15 = 30

Together one-hour work =  $\frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}$

So, P and Q together complete  $\frac{1}{6}$  of the work in 1 hour.

Time taken to complete the whole work = 6 hours

### 20. Option D

Length of Train P = 300 meters

Length of Train Q = 450 meters

Total distance to cross each other =  $300 + 450 = 750$  meters

Speed of Train P = 54 km/hr

Convert into m/sec:

Speed of Train P =  $54 \times \frac{5}{18} = 15$  m/sec

Speed of Train Q = 25 m/sec

When trains move in opposite directions, relative speed =  $15 + 25 = 40$  m/sec

Time taken in opposite directions =  $750/40$  seconds

When trains move in the same direction, relative speed =  $25 - 15 = 10$  m/sec

Time taken in the same direction =  $750/10$  seconds

Required ratio =  $750/40 : 750/10$

Cancel 750 from both sides.

Ratio =  $1/40 : 1/10$

Ratio =  $10 : 40 = 1 : 4$

Therefore, the required ratio is 1 : 4.

### 3. Quadratic Equations

## CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a)  $x > y$

(b)  $x < y$

(c)  $x = y$  or the relationship cannot be established

(d)  $x \geq y$

(e)  $x \leq y$

1.) I.  $12x^2 - 80x + 117 = 0$   
II.  $14y^2 + 40y - 136 = 0$

2.) I.  $7x^2 + 5x - 12 = 0$   
II.  $8y^2 - 46y + 51 = 0$

3.) I.  $5x^2 - 54x + 88 = 0$   
II.  $7y^2 + 5y - 38 = 0$

4.) I.  $12x^2 - 26x - 30 = 0$   
II.  $7y^2 - 67y + 138 = 0$

5.) I.  $8x^2 - 78x - 20 = 0$   
II.  $4y^2 - 52y + 144 = 0$

6.) I.  $7x^2 - 30x + 23 = 0$   
II.  $6y^2 + 19y + 14 = 0$

7.) I.  $14x^2 + 10x - 4 = 0$   
II.  $8y^2 - 76y + 120 = 0$

8.) I.  $12x^2 - 69x + 57 = 0$   
II.  $6y^2 + 3y - 9 = 0$

9.) I.  $10x^2 - 8x - 2 = 0$   
II.  $7y^2 - 75y + 68 = 0$

10.) I.  $8x^2 + 24x - 54 = 0$   
II.  $7y^2 - 60y + 53 = 0$

11.) I.  $8x^2 - 34x + 21 = 0$   
 II.  $12y^2 + 70y + 78 = 0$

12.) I.  $5x^2 + 9x - 14 = 0$   
 II.  $6y^2 - 65y + 106 = 0$

13.) I.  $5x^2 - 26x + 33 = 0$   
 II.  $10y^2 - 42y + 44 = 0$

14.) I.  $12x^2 + 14x - 10 = 0$   
 II.  $10y^2 - 41y + 18 = 0$

15.) I.  $8x^2 - 77x + 159 = 0$   
 II.  $14y^2 - 37y - 87 = 0$

16.) I.  $6x^2 - 61x + 10 = 0$   
 II.  $10y^2 + 19y + 7 = 0$

17.) I.  $14x^2 + 55x + 51 = 0$   
 II.  $9y^2 - 54y + 65 = 0$

18.) I.  $5x^2 - 37x + 66 = 0$   
 II.  $14y^2 - 8y - 102 = 0$

19.) I.  $6x^2 - 2x - 48 = 0$   
 II.  $4y^2 - 46y + 102 = 0$

20.) I.  $6x^2 - 34x - 112 = 0$   
 II.  $9y^2 + 30y - 24 = 0$

### Answers:

1. A
2. B
3. D
4. E
5. C
6. A
7. B
8. D
9. E
10. C
11. A
12. B
13. D
14. E
15. C
16. A
17. B
18. D
19. E
20. C

### Solutions:

$$(1) x = \frac{9}{2}, \frac{13}{6}$$

$$y = 2, -\frac{34}{7}$$

$$(2) x = 1, -\frac{12}{7}$$

$$y = \frac{17}{4}, \frac{3}{2}$$

$$(3) x = \frac{44}{5}, 2$$

$$y = 2, -\frac{19}{7}$$

$$(4) x = 3, -\frac{5}{6}$$

$$y = \frac{46}{7}, 3$$

$$(5) x = 10, -\frac{1}{4}$$

$$y = 9, 4$$

$$(6) x = \frac{23}{7}, 1$$

$$y = -\frac{7}{6}, -2$$

$$(7) x = \frac{2}{7}, -1$$

$$y = \frac{15}{2}, 2$$

$$(8) x = \frac{19}{4}, 1$$

$$y = 1, -\frac{3}{2}$$

$$(9) x = 1, -\frac{1}{5}$$

$$y = \frac{68}{7}, 1$$

$$(10) x = \frac{3}{2}, -\frac{9}{2}$$

$$y = \frac{53}{7}, 1$$

$$(11) x = \frac{7}{2}, \frac{3}{4}$$

$$y = -\frac{3}{2}, -\frac{13}{3}$$

$$(12) x = 1, -\frac{14}{5}$$

$$y = \frac{53}{6}, 2$$

$$(13) x = 3, \frac{11}{5}$$

$$y = \frac{11}{5}, 2$$

$$(14) x = \frac{1}{2}, -\frac{5}{3}$$

$$y = \frac{18}{5}, \frac{1}{2}$$

$$(15) x = \frac{53}{8}, 3$$

$$y = \frac{29}{7}, -\frac{3}{2}$$

$$(16) x = 10, \frac{1}{6}$$

$$y = -\frac{1}{2}, \frac{7}{5}$$

$$(17) x = -\frac{3}{2}, -\frac{17}{7}$$

$$y = \frac{13}{3}, \frac{5}{3}$$

$$(18) x = \frac{22}{5}, 3$$

$$y = 3, -\frac{17}{7}$$

$$(19) x = 3, -\frac{8}{3}$$

$$y = \frac{17}{2}, 3$$

$$(20) x = 8, -\frac{7}{3}$$

$$y = \frac{2}{3}, -4$$

## 4. WRONG NUMBER SERIES

(1) 1840, 1858, 1887, 1915, 1954, 2000

- (a) 1858
- (b) 1954
- (c) 1887
- (d) 2000
- (e) None of these

(2) 126, 257, 781, 3149, 15715, 94315

- (a) 257
- (b) 781
- (c) 15715
- (d) 3149
- (e) None of these

(3) 4096, 2050, 1004, 478, 207, 63.5

- (a) 2050
- (b) 1004
- (c) 478
- (d) 63.5
- (e) None of these

(4) 2500, 2456, 2404, 2336, 2250, 2168

- (a) 2404
- (b) 2250
- (c) 2168
- (d) 2456
- (e) None of these

(5) 180, 353, 692, 1363, 2708, 5361

- (a) 353
- (b) 692
- (c) 1363
- (d) 5361
- (e) None of these

(6) 980, 1005, 985, 1010, 990, 1035

- (a) 1010
- (b) 985
- (c) 1005
- (d) 990
- (e) None of these

(7) 740, 761, 793, 838, 898, 965

- (a) 761
- (b) 838



- |                                              |                                     |
|----------------------------------------------|-------------------------------------|
| (c) 793                                      | (a) 748                             |
| (d) 965                                      | (b) 968                             |
| (e) None of these                            | (c) 828                             |
|                                              | (d) 1230                            |
| (8) 112, 215, 635, 2481, 12369, 74169        | (e) None of these                   |
| (a) 112                                      |                                     |
| (b) 635                                      | (12) 120, 890, 250, 1780, 510, 3540 |
| (c) 2481                                     | (a) 890                             |
| (d) 74169                                    | (b) 250                             |
| (e) None of these                            | (c) 1780                            |
|                                              | (d) 510                             |
| (9) 640, 648, 672, 720, 810, 920             | (e) None of these                   |
| (a) 648                                      |                                     |
| (b) 672                                      | (13) 856, 870, 895, 942, 1033, 1222 |
| (c) 720                                      | (a) 870                             |
| (d) 920                                      | (b) 942                             |
| (e) None of these                            | (c) 895                             |
|                                              | (d) 1033                            |
| (10) 450.5, 457.75, 472.25, 489, 523, 559.25 | (e) None of these                   |
| (a) 450.5                                    | (14) 600, 650, 702, 766, 812, 870   |
| (b) 457.75                                   | (a) 702                             |
| (c) 489                                      | (b) 870                             |
| (d) 523                                      | (c) 766                             |
| (e) None of these                            | (d) 650                             |
|                                              | (e) None of these                   |
| (11) 700, 712, 748, 828, 968, 1230           |                                     |

(15) 310, 303, 909, 895, 3560, 3559

(a) 303

(b) 909

(c) 895

(d) 3560

(e) None of these

(16) 1150, 1171, 1206, 1257, 1336, 1415

(a) 1336

(b) 1206

(c) 1171

(d) 1415

(e) None of these

(17) 45, 276, 1375, 5544, 16635, 33272

(a) 276

(b) 1375

(c) 5544

(d) 33272

(e) None of these

(18) 880, 892.5, 867.5, 905, 855, 927.5

(a) 892.5

(b) 867.5

(c) 855

(d) 927.5

(e) None of these

(19) 1800, 1782, 1760, 1710, 1638, 1521

(a) 1710

(b) 1638

(c) 1760

(d) 1521

(e) None of these

(20) 5520, 2790, 970, 300, 118.5, 89.75

(a) 2790

(b) 970

(c) 118.5

(d) 89.75

(e) None of these

**Answers:**

(1) c

(2) d

(3) a

(4) b

(5) e

(6) a

(7) d

(8) b

- |        |                                                                                               |
|--------|-----------------------------------------------------------------------------------------------|
| (9) e  | (7) $+(3 \times 7)$ , $+(4 \times 8)$ , $+(5 \times 9)$ , $+(6 \times 10)$ , $+(7 \times 11)$ |
| (10) c |                                                                                               |
| (11) b | (8) $\times 2 - 9$ , $\times 3 - 18$ , $\times 4 - 27$ , $\times 5 - 36$ , $\times 6 - 45$    |
| (12) a | (9) $+(1 \times 8)$ , $+(3 \times 8)$ , $+(6 \times 8)$ , $+(10 \times 8)$ , $+(15 \times 8)$ |
| (13) e |                                                                                               |
| (14) c | (10) $+7.25$ , $+14.5$ , $+21.75$ , $+29$ , $+36.25$                                          |
| (15) d | (11) $+(2^2 + 2^3)$ , $+(3^2 + 3^3)$ , $+(4^2 + 4^3)$ , $+(5^2 + 5^3)$ , $+(6^2 + 6^3)$       |
| (16) a |                                                                                               |
| (17) b | (12) Odd positions: $\times 2 + 10$ ; Even positions: $\times 2 - 20$                         |
| (18) d |                                                                                               |
| (19) c | (13) $+14$ , $+25$ , $+47$ , $+91$ , $+179$ ; each next difference = previous difference      |
| (20) e | $\times 2 - 3$                                                                                |

**Solutions:**

- |                                                                                                  |                                                                                                          |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| (1) $+18$ , $+25$ , $+32$ , $+39$ , $+46$                                                        | (14) $25 \times 24$ , $26 \times 25$ , $27 \times 26$ , $28 \times 27$ , $29 \times 28$ , $30 \times 29$ |
| (2) $\times 2 + 5$ , $\times 3 + 10$ , $\times 4 + 15$ , $\times 5 + 20$ , $\times 6 + 25$       | (15) $-7$ , $\times 3$ , $-14$ , $\times 4$ , $-21$                                                      |
| (3) $\div 2 - 8$ , $\div 2 - 16$ , $\div 2 - 24$ , $\div 2 - 32$ , $\div 2 - 40$                 | (16) $+21$ , $+35$ , $+51$ , $+69$ , $+89$ ; second differences: $+14$ , $+16$ , $+18$ , $+20$           |
| (4) $-(11 \times 4)$ , $-(13 \times 4)$ , $-(17 \times 4)$ , $-(19 \times 4)$ , $-(23 \times 4)$ | (17) $\times 6 + 6$ , $\times 5 + 5$ , $\times 4 + 4$ , $\times 3 + 3$ , $\times 2 + 2$                  |
| (5) $\times 2 - 7$ , $\times 2 - 14$ , $\times 2 - 21$ , $\times 2 - 28$ , $\times 2 - 35$       | (18) $+12.5$ , $-25$ , $+37.5$ , $-50$ , $+62.5$                                                         |
| (6) $+25$ , $-20$ , $+35$ , $-30$ , $+45$                                                        | (19) $-18$ , $-27$ , $-45$ , $-72$ , $-117$ ; each next difference = sum of previous two differences     |
|                                                                                                  | (20) $\div 2 + 30$ , $\div 3 + 40$ , $\div 4 + 50$ , $\div 5 + 60$ , $\div 6 + 70$                       |

## 5. MISSING NUMBER SERIES

(1) 4725, 4706, 4668, 4611, ?, 4440

- (a) 4531
- (b) 4533
- (c) 4535
- (d) 4537
- (e) None of these

(2) 6, 14, 30, ?, 94, 146

- (a) 54
- (b) 56
- (c) 58
- (d) 60
- (e) None of these

(3) 28, 82, 242, ?, 2152, 6446

- (a) 720
- (b) 716
- (c) 724
- (d) 728
- (e) None of these

(4) 142, 219, 368, ?, 969, 1570

- (a) 590
- (b) 592
- (c) 594
- (d) 596
- (e) None of these

(5) 1320, 675, 240, ?, 30, 20

- (a) 73
- (b) 75
- (c) 77

(d) 79

(e) None of these

(6) 43.5, 50, 58.5, ?, 81.5, 96

- (a) 68
- (b) 69
- (c) 70
- (d) 71

(e) None of these

(7) 282, 295, 321, ?, 412, 477

- (a) 360
- (b) 358
- (c) 356
- (d) 362

(e) None of these

(8) 6, 17, 47, ?, 380, 1115

- (a) 132
- (b) 130
- (c) 134
- (d) 136

(e) None of these

(9) 2048, 2039, 2014, ?, 1884, 1763

- (a) 1959
- (b) 1961
- (c) 1963
- (d) 1965

(e) None of these

(10) 14, 38, 82, ?, 254, 394

- (a) 152
- (b) 150
- (c) 148
- (d) 154
- (e) None of these

(11) 2500, 500, 2000, 400, ?, 320

- (a) 1580
- (b) 1590
- (c) 1610
- (d) 1600
- (e) None of these

(12) 9, 14, 26, ?, 78, 132

- (a) 43
- (b) 45
- (c) 47
- (d) 49
- (e) None of these

(13) 55, 73, 104, ?, 213, 295

- (a) 150
- (b) 148
- (c) 146
- (d) 152
- (e) None of these

(14) 848, 432, 224, ?, 68, 42

- (a) 116
- (b) 118
- (c) 122
- (d) 120
- (e) None of these

(15) 3900, 3868, 3823, ?, 3686, 3590

- (a) 3761
- (b) 3763
- (c) 3765
- (d) 3767
- (e) None of these

(16) 26, 59, 129, 271, ?, 1137

- (a) 553
- (b) 555
- (c) 557
- (d) 559
- (e) None of these

(17) 5, 18, 49, ?, 197, 330

- (a) 102
- (b) 104
- (c) 106
- (d) 108
- (e) None of these

(18) 560, 584, 624, ?, 768, 880

- (a) 680
- (b) 682
- (c) 684
- (d) 686
- (e) None of these

(19) 5000, 4992, 4965, ?, 4776, 4560

- (a) 4897
- (b) 4899
- (c) 4901
- (d) 4903
- (e) None of these

(20) 12, 23, 67, ?, 1321, 7921

- (a) 261
- (b) 263
- (c) 265
- (d) 267
- (e) None of these

### Answers:

- (1) c
- (2) b

- (3) a  
 (4) c  
 (5) b  
 (6) b  
 (7) a  
 (8) a  
 (9) d  
 (10) a  
 (11) d  
 (12) b  
 (13) a  
 (14) d  
 (15) b  
 (16) d  
 (17) c  
 (18) c  
 (19) c  
 (20) c

**Solutions:**

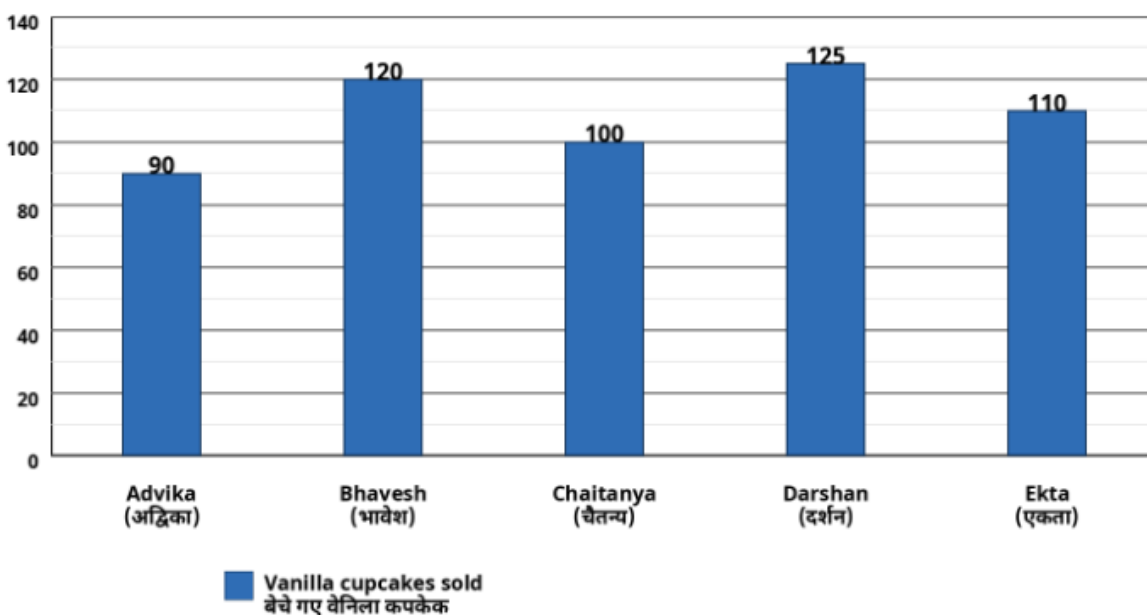
- (1)  $-19 \times 1, -19 \times 2, -19 \times 3, -19 \times 4, -19 \times 5$   
 (2)  $+8, +16, +26, +38, +52$   
 Second difference:  $+8, +10, +12, +14$   
 (3)  $\times 3 - 2, \times 3 - 4, \times 3 - 6, \times 3 - 8, \times 3 - 10$   
 (4) Sum of previous two numbers  $+ 7$   
 (5)  $\div 2 + 15, \div 3 + 15, \div 4 + 15, \div 5 + 15, \div 6 + 15$

- (6)  $+6.5, +8.5, +10.5, +12.5, +14.5$   
 (7)  $+13, +26, +39, +52, +65$   
 (8)  $\times 3 - 1^2, \times 3 - 2^2, \times 3 - 3^2, \times 3 - 4^2, \times 3 - 5^2$   
 (9)  $-3^2, -5^2, -7^2, -9^2, -11^2$   
 (10) Terms are:  $n^3 + n^2 + 2$ , for  $n = 2, 3, 4, 5, 6, 7$   
 (11)  $\div 5, \times 4, \div 5, \times 4, \div 5$   
 (12) Sum of previous two numbers  $+3, +5, +7, +9$   
 (13)  $+18, +31, +46, +63, +82$   
 Second difference:  $+13, +15, +17, +19$   
 (14)  $\div 2 + 8, \div 2 + 8, \div 2 + 8, \div 2 + 8, \div 2 + 8$   
 (15)  $-4 \times 8, -5 \times 9, -6 \times 10, -7 \times 11, -8 \times 12$   
 (16)  $\times 2 + 7, \times 2 + 11, \times 2 + 13, \times 2 + 17, \times 2 + 19$   
 (17)  $+13, +31, +57, +91, +133$   
 Second difference:  $+18, +26, +34, +42$   
 (18)  $+4 \times 6, +5 \times 8, +6 \times 10, +7 \times 12, +8 \times 14$   
 (19)  $-2^3, -3^3, -4^3, -5^3, -6^3$   
 (20)  $\times 2 - 1, \times 3 - 2, \times 4 - 3, \times 5 - 4, \times 6 - 5$

## 6. DATA INTERPRETATION

**Set 1:** The bar graph shows the number of Vanilla cupcakes sold by five different bakery counters. Read the data carefully and answer the following questions.

दंड आलेख पाँच अलग-अलग बेकरी काउंटरों द्वारा बेचे गए Vanilla cupcakes की संख्या दर्शाता है। डेटा को ध्यानपूर्वक पढ़ें और निम्नलिखित प्रश्नों के उत्तर दें। डेटा तालिका के सभी मान 5 के गुणज हैं।



Q1: If the number of Chocolate cupcakes sold by Bhavesh is  $37\frac{1}{2}\%$  more than the number of Vanilla cupcakes sold by Bhavesh, and the number of Chocolate cupcakes sold by Chaitanya is 20 more than the number of Vanilla cupcakes sold by Darshan, then find the total number of Chocolate cupcakes sold by Bhavesh and Chaitanya together.

यदि भावेश द्वारा बेचे गए Chocolate cupcakes की संख्या, भावेश द्वारा बेचे गए Vanilla cupcakes से  $37\frac{1}{2}\%$  अधिक है, और चैतन्य द्वारा बेचे गए Chocolate cupcakes की संख्या, दर्शन द्वारा बेचे गए Vanilla cupcakes से 20 अधिक है, तो भावेश और चैतन्य द्वारा मिलाकर बेचे गए Chocolate cupcakes की कुल संख्या ज्ञात कीजिए।

- (A) 295
- (B) 300
- (C) 310

- (D) 325
- (E) 335

Q2: Find the average number of Vanilla cupcakes sold by Ekta, Bhavesh and Chaitanya.

एकता, भावेश और चैतन्य द्वारा बेचे गए Vanilla cupcakes की औसत संख्या ज्ञात कीजिए।

- (A) 110
- (B) 105
- (C) 115
- (D) 120
- (E) 125

Q3: If the ratio of Chocolate cupcakes sold by Advika to Chocolate cupcakes sold by Bhavesh is 2:1, and the ratio of Chocolate cupcakes sold by Bhavesh to Vanilla cupcakes sold by Chaitanya is 1:2, then find the number of Chocolate cupcakes sold by Advika.

यदि अद्विका और भावेश द्वारा बेचे गए Chocolate cupcakes का अनुपात 2:1 है, और भावेश द्वारा बेचे गए Chocolate cupcakes तथा चैतन्य द्वारा बेचे गए Vanilla cupcakes का अनुपात 1:2 है, तो अद्विका द्वारा बेचे गए Chocolate cupcakes की संख्या ज्ञात कीजिए।

- (A) 80
- (B) 90
- (C) 95
- (D) 105
- (E) 100

Q4: Find the difference between the number of Vanilla cupcakes sold by Advika and Ekta.

अद्विका और एकता द्वारा बेचे गए Vanilla cupcakes की संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 15
- (B) 20
- (C) 25
- (D) 30
- (E) 35

Q5: Vanilla cupcakes sold by Chaitanya are what percentage of Vanilla cupcakes sold by Darshan?

चैतन्य द्वारा बेचे गए Vanilla cupcakes, दर्शन द्वारा बेचे गए Vanilla cupcakes का कितना



प्रतिशत हैं?

- (A) 60%
- (B) 70%
- (C) 75%
- (D) 80%
- (E) 85%

**Solutions:**

1. Option C

Vanilla cupcakes sold by Bhavesh = 120

Chocolate cupcakes sold by Bhavesh =  $120 \times \frac{11}{8} = 165$

Vanilla cupcakes sold by Darshan = 125

Chocolate cupcakes sold by Chaitanya =  $125 + 20 = 145$

Total Chocolate cupcakes =  $165 + 145 = 310$

2. Option A

Vanilla cupcakes sold by Ekta, Bhavesh and Chaitanya = 110, 120 and 100

Average =  $(110 + 120 + 100) / 3$

Average =  $330 / 3 = 110$

3. Option E

Vanilla cupcakes sold by Chaitanya = 100

Chocolate cupcakes sold by Bhavesh =  $100 \times \frac{1}{2} = 50$

Chocolate cupcakes sold by Advika =  $50 \times 2 = 100$

4. Option B

Vanilla cupcakes sold by Ekta = 110

Vanilla cupcakes sold by Advika = 90

Difference =  $110 - 90 = 20$

5. Option D

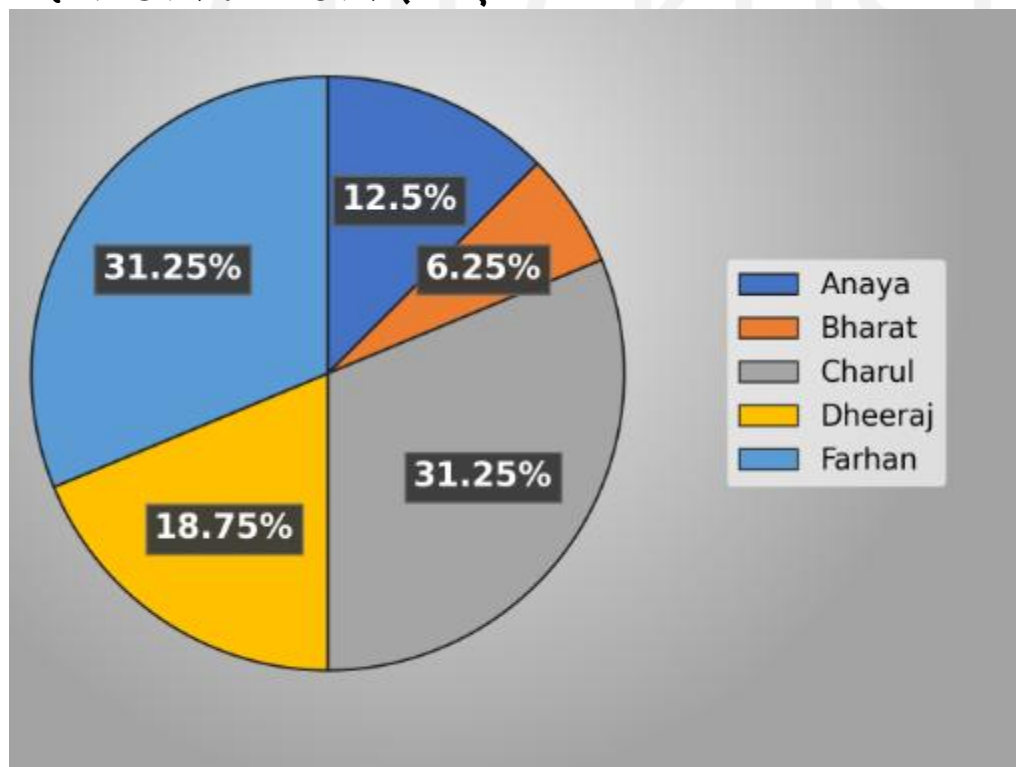
Vanilla cupcakes sold by Chaitanya = 100

Vanilla cupcakes sold by Darshan = 125

Required percentage =  $100 / 125 \times 100 = 80\%$

**Set 2:** The pie chart shows the percentage distribution of the total number of reusable bottles, steel and copper together, sold by five different outlets: Anaya, Bharat, Charul, Dheeraj and Farhan. The total number of reusable bottles sold by all five outlets together is 4800. Read the pie chart carefully and answer the following questions.

पाई चार्ट पाँच अलग-अलग आउटलेट, अनया, भारत, चारुल, धीरज और फ़रहान द्वारा बेची गई कुल पुनः उपयोग योग्य बोतलों, यानी स्टील और कॉपर बोतलों को मिलाकर, का प्रतिशत वितरण दर्शाता है। सभी पाँच आउटलेट द्वारा मिलाकर बेची गई कुल पुनः उपयोग योग्य बोतलों की संख्या 4800 है। पाई चार्ट को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



**Q1:** Find the difference between the total bottles sold by Dheeraj and the total bottles sold by Bharat.

धीरज द्वारा बेची गई कुल बोतलों और भारत द्वारा बेची गई कुल बोतलों के बीच का अंतर ज्ञात कीजिए।

- (A) 540
- (B) 720
- (C) 660
- (D) 600
- (E) 480

Q2: If the number of steel bottles sold by Farhan is  $8\frac{1}{3}\%$  more than the number of copper bottles sold by the same outlet, then find the number of copper bottles sold by Farhan as a percentage of the total bottles sold by Anaya.

यदि फ़रहान द्वारा बेची गई स्टील बोतलों की संख्या, उसी आउटलेट द्वारा बेची गई कॉपर बोतलों की संख्या से  $8\frac{1}{3}\%$  अधिक है, तो फ़रहान द्वारा बेची गई कॉपर बोतलों, अनया द्वारा बेची गई कुल बोतलों का कितना प्रतिशत हैं?

- (A) 110%
- (B) 120%
- (C) 125%
- (D) 100%
- (E) 135%

Q3: Find the ratio of total bottles sold by Charul to the total bottles sold by Anaya and Bharat together.

चारुल द्वारा बेची गई कुल बोतलों का अनया और भारत द्वारा मिलाकर बेची गई कुल बोतलों से अनुपात ज्ञात कीजिए।

- (A) 7:4
- (B) 4:3
- (C) 3:2
- (D) 8:5
- (E) 5:3

Q4: If the total bottles sold by a new outlet, Nilesh, are 180 more than the total bottles sold by Dheeraj, then find the average number of bottles sold by Nilesh and Farhan.

यदि एक नए आउटलेट, नीलेश, द्वारा बेची गई कुल बोतलों धीरज द्वारा बेची गई कुल बोतलों से 180 अधिक हैं, तो नीलेश और फ़रहान द्वारा बेची गई बोतलों की औसत संख्या ज्ञात कीजिए।

- (A) 1290
- (B) 1170
- (C) 1230
- (D) 1350
- (E) 1410

Q5: Find the corresponding central angle, in degrees, for the total bottles sold by Bharat and Charul together.

भारत और चारुल द्वारा मिलाकर बेची गई कुल बोतलों के लिए संबंधित केंद्रीय कोण, डिग्री

में, ज्ञात कीजिए।

- (A) 120  
(B) 150  
(C) 135  
(D) 108  
(E) 144

**Solutions:**

| Outlet  | Percentage | Calculation             | Total bottles sold |
|---------|------------|-------------------------|--------------------|
| Anaya   | 12.5%      | $12.5/100 \times 4800$  | 600                |
| Bharat  | 6.25%      | $6.25/100 \times 4800$  | 300                |
| Charul  | 31.25%     | $31.25/100 \times 4800$ | 1500               |
| Dheeraj | 18.75%     | $18.75/100 \times 4800$ | 900                |
| Farhan  | 31.25%     | $31.25/100 \times 4800$ | 1500               |

Common Calculation:

Total bottles sold = 4800

Anaya =  $12 \frac{1}{2}\%$  of 4800 = 600

Bharat =  $6 \frac{1}{4}\%$  of 4800 = 300

Charul =  $31 \frac{1}{4}\%$  of 4800 = 1500

Dheeraj =  $18 \frac{3}{4}\%$  of 4800 = 900

Farhan =  $31 \frac{1}{4}\%$  of 4800 = 1500

1. Option D

Dheeraj sold = 900

Bharat sold = 300

Required difference =  $900 - 300 = 600$

2. Option B

Steel bottles are  $8 \frac{1}{3}\%$  more than copper bottles

So, steel:copper = 13:12

Farhan's total bottles = 1500

Copper bottles sold by Farhan =  $1500 \times \frac{12}{25} = 720$

Anaya's total bottles = 600

Required percentage =  $\frac{720}{600} \times 100 = 120\%$

3. Option E

Charul sold = 1500

Anaya + Bharat =  $600 + 300 = 900$

Required ratio =  $1500:900 = 5:3$

4. Option A

Dheeraj sold = 900

Nilesh sold =  $900 + 180 = 1080$

Farhan sold = 1500

Required average =  $(1080 + 1500)/2 = 1290$

5. Option C

Bharat + Charul =  $6\frac{1}{4}\% + 31\frac{1}{4}\% = 37\frac{1}{2}\%$

Central angle =  $37\frac{1}{2}\%$  of 360

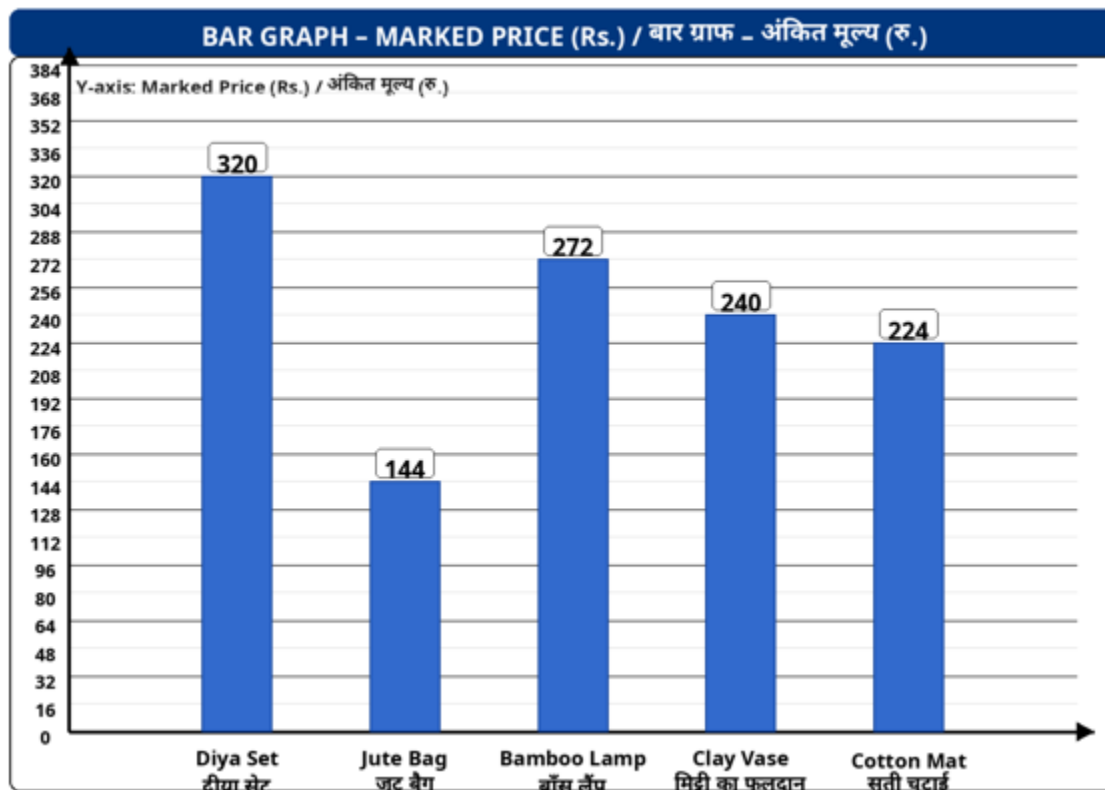
Central angle = 135 degrees

**Set 3:** The bar graph shows the marked price (in Rs.) of five different handicraft products sold in a fair. The table shows the difference (in Rs.) between the selling price (SP) and the cost price (CP) of each product.

बार ग्राफ मेले में बेचे जाने वाले पाँच अलग-अलग हस्तशिल्प उत्पादों के अंकित मूल्य (रुपये में) को दिखाता है। तालिका प्रत्येक उत्पाद के विक्रय मूल्य (SP) और क्रय मूल्य (CP) के बीच का अंतर (रुपये में) दिखाती है।

Note: Positive SP-CP means profit, and negative SP-CP means loss.

नोट: धनात्मक SP-CP लाभ को दर्शाता है और ऋणात्मक SP-CP हानि को दर्शाता है।



| Product / उत्पाद             | SP-CP (Rs.) / SP-CP (रु.) |
|------------------------------|---------------------------|
| Diya Set / दीया सेट          | -32                       |
| Jute Bag / जूट बैग           | +24                       |
| Bamboo Lamp / बाँस लैंप      | +48                       |
| Clay Vase / मिट्टी का फूलदान | -64                       |
| Cotton Mat / सूती चटाई       | +56                       |

Q1: Find the number of products whose marked price is less than the average marked price of all five products together.

उन उत्पादों की संख्या ज्ञात कीजिए जिनका अंकित मूल्य सभी पाँच उत्पादों के औसत अंकित मूल्य से कम है।

- (A) 5
- (B) 4
- (C) 2
- (D) 3
- (E) 1

Q2: Find the ratio of the combined marked price of the Diya Set and Cotton Mat to the combined marked price of the Bamboo Lamp and Jute Bag. If the ratio is  $p:q$  in simplest form, find  $p \times q$ .

दीया सेट और सूती चटाई के मिलाकर अंकित मूल्य का, बाँस लैंप और जूट बैग के मिलाकर अंकित मूल्य से अनुपात ज्ञात कीजिए। यदि सरलतम रूप में अनुपात  $p:q$  है, तो  $p \times q$  ज्ञात कीजिए।

- (A) 187
- (B) 209
- (C) 195
- (D) 247
- (E) 221

Q3: If the cost price of the Diya Set is equal to the marked price of the Clay Vase, find the discount percentage applied to the Diya Set.

यदि दीया सेट का क्रय मूल्य, मिट्टी के फूलदान के अंकित मूल्य के बराबर है, तो दीया सेट पर दी गई छूट प्रतिशत ज्ञात कीजिए।

- (A) 35%
- (B) 28%
- (C) 42%
- (D) 32%
- (E) 38%

Q4: If the discount percentages applied to the Jute Bag and Bamboo Lamp are  $16\frac{2}{3}\%$  and  $12\frac{1}{2}\%$ , respectively, find the sum of their cost prices. (in Rs.)

यदि जूट बैग और बाँस लैंप पर दी गई छूट क्रमशः  $16\frac{2}{3}\%$  और  $12\frac{1}{2}\%$  है, तो दोनों के क्रय मूल्यों का योग ज्ञात कीजिए। (रुपये में)

- (A) 274

- (B) 296  
(C) 268  
(D) 286  
(E) 304

Q5: If the ratio of the selling price of the Bamboo Lamp to that of the Clay Vase is 15:8 and the cost prices of the Bamboo Lamp and Clay Vase are Rs. X each, find 2X.

यदि बाँस लैंप और मिट्टी के फूलदान के विक्रय मूल्यों का अनुपात 15:8 है और बाँस लैंप तथा मिट्टी के फूलदान के क्रय मूल्य प्रत्येक Rs. X हैं, तो 2X ज्ञात कीजिए।

- (A) 368  
(B) 384  
(C) 392  
(D) 360  
(E) 376

Solutions:

| Product / उत्पाद             | Marked Price / अंकित मूल्य | SP-CP / SP-CP | Selling Price / विक्रय मूल्य | Cost Price / क्रय मूल्य |
|------------------------------|----------------------------|---------------|------------------------------|-------------------------|
| Diya Set / दीया सेट          | 320                        | -32           | 208                          | 240                     |
| Jute Bag / जूट बैग           | 144                        | +24           | 120                          | 96                      |
| Bamboo Lamp / बाँस लैंप      | 272                        | +48           | 238                          | 190                     |
| Clay Vase / मिट्टी का फूलदान | 240                        | -64           | 128                          | 192                     |
| Cotton Mat / सूती चटाई       | 224                        | +56           | 224                          | 168                     |

1. Option C

Average marked price =  $(320 + 144 + 272 + 240 + 224)/5 = 240$

Products below 240 = Jute Bag and Cotton Mat

Required number = 2

2. Option E

Required ratio =  $(320 + 224):(272 + 144)$

Required ratio =  $544:416 = 17:13$

$p \times q = 17 \times 13 = 221$

3. Option A

CP of Diya Set = MP of Clay Vase = 240

SP-CP for Diya Set = -32, so SP =  $240 - 32 = 208$

Discount =  $320 - 208 = 112$



$$\text{Discount\%} = 112/320 \times 100 = 35\%$$

#### 4. Option D

$$\text{SP of Jute Bag} = 144 \times 83 \frac{1}{3}\% = 144 \times \frac{5}{6} = 120$$

$$\text{CP of Jute Bag} = 120 - 24 = 96$$

$$\text{SP of Bamboo Lamp} = 272 \times 87 \frac{1}{2}\% = 272 \times \frac{7}{8} = 238$$

$$\text{CP of Bamboo Lamp} = 238 - 48 = 190$$

$$\text{Required sum} = 96 + 190 = 286$$

#### 5. Option B

$$\text{Let CP of Bamboo Lamp} = \text{CP of Clay Vase} = X$$

$$\text{SP of Bamboo Lamp} = X + 48$$

$$\text{SP of Clay Vase} = X - 64$$

$$(X + 48):(X - 64) = 15:8$$

$$8X + 384 = 15X - 960$$

$$7X = 1344, \text{ so } X = 192$$

$$2X = 384$$

**Set 4:** The given data gives information about the number of scholarship applications processed by three coaching centres, Ujjwal Centre, Prerna Centre and Sarthak Centre, in two different years 2022 and 2023. The ratio of applications processed by Ujjwal Centre in 2022 to applications processed by Sarthak Centre in 2023 is 5 : 6. In 2023, the number of applications processed by Prerna Centre is equal to the number of applications processed by Ujjwal Centre in 2022. The number of applications processed by Ujjwal Centre in 2023 is 25% more than the number of applications processed by Prerna Centre in 2022. The number of applications processed by Sarthak Centre in 2022 is 25% less than the number of applications processed by Prerna Centre in 2022. The total number of applications processed by all three centres in 2022 is 558, and the total number of applications processed by all three centres in 2023 is 666.

दिया गया डेटा तीन कोचिंग केंद्रों, उज्ज्वल केंद्र, प्रेरणा केंद्र और सार्थक केंद्र, द्वारा दो अलग-अलग वर्षों 2022 और 2023 में प्रोसेस किए गए स्कॉलरशिप आवेदनों की संख्या से संबंधित है। 2022 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों और 2023 में सार्थक केंद्र द्वारा प्रोसेस किए गए आवेदनों का अनुपात 5 : 6 है। 2023 में प्रेरणा केंद्र द्वारा प्रोसेस किए गए आवेदनों की संख्या, 2022 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों की संख्या के बराबर है। 2023 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों की संख्या, 2022 में प्रेरणा केंद्र द्वारा प्रोसेस किए गए आवेदनों की संख्या से 25% अधिक है। 2022 में सार्थक केंद्र द्वारा प्रोसेस किए गए आवेदनों की संख्या, 2022 में प्रेरणा केंद्र द्वारा प्रोसेस किए गए आवेदनों की संख्या से 25% कम है। 2022 में तीनों केंद्रों द्वारा प्रोसेस किए गए कुल आवेदनों की संख्या 558 है, और 2023 में तीनों केंद्रों द्वारा प्रोसेस किए गए कुल आवेदनों की संख्या 666 है।

Q1: Applications processed by Sarthak Centre in 2023 are what percentage of applications processed by Prerna Centre in 2022?

2023 में सार्थक केंद्र द्वारा प्रोसेस किए गए आवेदन, 2022 में प्रेरणा केंद्र द्वारा प्रोसेस किए गए आवेदनों का कितना प्रतिशत हैं?

- (A) 80%
- (B) 120%
- (C) 90%
- (D) 100%
- (E) 110%

Q2: If applications processed by Tarun Centre in 2023 are 20% less than the applications processed by Ujjwal Centre in 2022, then find the total number of applications processed by Ujjwal Centre, Prerna Centre, Sarthak Centre and Tarun

Centre together in 2023.

यदि 2023 में तरुण केंद्र द्वारा प्रोसेस किए गए आवेदन, 2022 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों से 20% कम हैं, तो 2023 में उज्ज्वल केंद्र, प्रेरणा केंद्र, सार्थक केंद्र और तरुण केंद्र द्वारा मिलाकर प्रोसेस किए गए कुल आवेदनों की संख्या ज्ञात कीजिए।

- (A) 792
- (B) 810
- (C) 828
- (D) 846
- (E) 774

Q3: Find the difference between the average number of applications processed by Sarthak Centre and Prerna Centre in both the years.

दोनों वर्षों में सार्थक केंद्र और प्रेरणा केंद्र द्वारा प्रोसेस किए गए आवेदनों की औसत संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 18
- (B) 27
- (C) 36
- (D) 45
- (E) 9

Q4: If the ratio of applications processed by Ujjwal Centre in 2024 to applications processed by Ujjwal Centre in 2023 is 4 : 5, then applications processed by Sarthak Centre in 2022 are what percentage more or less than applications processed by Ujjwal Centre in 2024?

यदि 2024 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों और 2023 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों का अनुपात 4 : 5 है, तो 2022 में सार्थक केंद्र द्वारा प्रोसेस किए गए आवेदन, 2024 में उज्ज्वल केंद्र द्वारा प्रोसेस किए गए आवेदनों से कितने प्रतिशत अधिक या कम हैं?

- (A) 25% less
- (B) 20% more
- (C) 15% less
- (D) 30% more
- (E) 10% less

Q5: In 2023, 30% of the applications processed by Prerna Centre and 25% of the applications processed by Sarthak Centre were incomplete. Find the total number of complete applications processed by Prerna Centre and Sarthak Centre together

in 2023.

2023 में प्रेरणा केंद्र द्वारा प्रोसेस किए गए आवेदनों में से 30% और सार्थक केंद्र द्वारा प्रोसेस किए गए आवेदनों में से 25% अधूरे थे। 2023 में प्रेरणा केंद्र और सार्थक केंद्र द्वारा मिलाकर प्रोसेस किए गए पूर्ण आवेदनों की कुल संख्या ज्ञात कीजिए।

- (A) 270
- (B) 306
- (C) 288
- (D) 324
- (E) 252

**Solutions:**

| Centre                | 2022       | 2023       | Total       |
|-----------------------|------------|------------|-------------|
| <b>Ujjwal Centre</b>  | 180        | 270        | 450         |
| <b>Prerna Centre</b>  | 216        | 180        | 396         |
| <b>Sarthak Centre</b> | 162        | 216        | 378         |
| <b>Total</b>          | <b>558</b> | <b>666</b> | <b>1224</b> |

1. Option D

Applications processed by Sarthak Centre in 2023 = 216

Applications processed by Prerna Centre in 2022 = 216

Required percentage =  $216 \div 216 \times 100$

= 100%

2. Option B

Applications processed by Ujjwal Centre in 2022 = 180

Applications processed by Tarun Centre in 2023 = 80% of 180

= 144

Total applications processed by three centres in 2023 = 666

Required total = 666 + 144

= 810

## 3. Option E

Average applications processed by Sarthak Centre =  $(162 + 216) \div 2$   
= 189

Average applications processed by Prerna Centre =  $(216 + 180) \div 2$   
= 198

Required difference =  $198 - 189$   
= 9

## 4. Option A

Applications processed by Ujjwal Centre in 2023 = 270

Ujjwal Centre in 2024 : Ujjwal Centre in 2023 = 4 : 5

Applications processed by Ujjwal Centre in 2024 =  $270 \times 4 \div 5$   
= 216

Applications processed by Sarthak Centre in 2022 = 162

Required percentage =  $(216 - 162) \div 216 \times 100$   
= 25% less

## 5. Option C

Applications processed by Prerna Centre in 2023 = 180

Complete applications of Prerna Centre = 70% of 180 = 126

Applications processed by Sarthak Centre in 2023 = 216

Complete applications of Sarthak Centre = 75% of 216 = 162

Required total =  $126 + 162$   
= 288